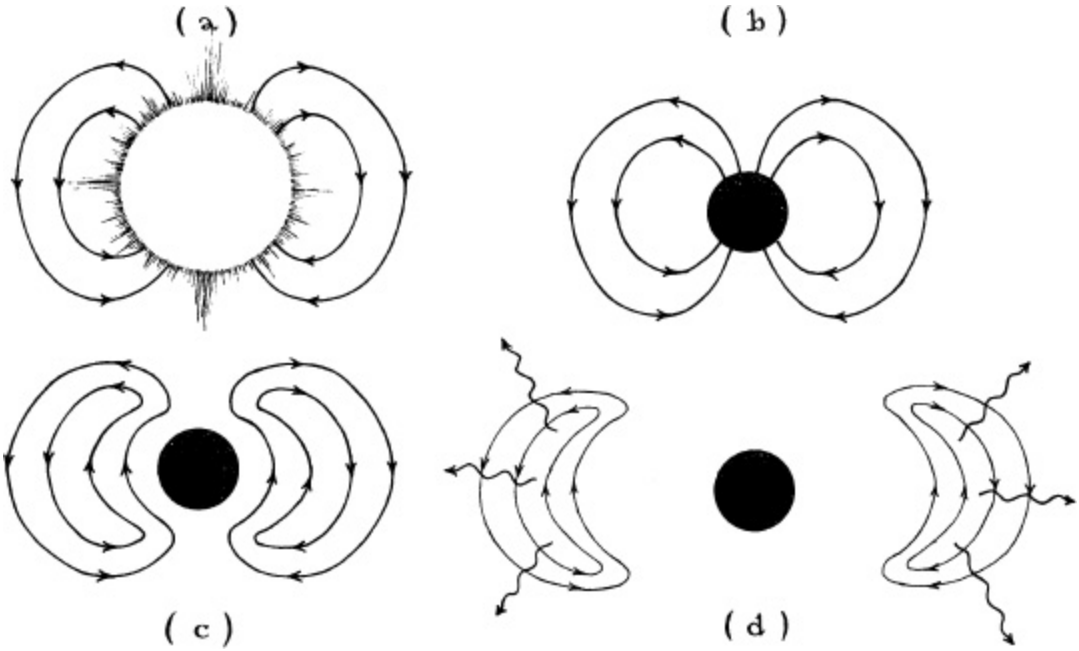


7.5 A sequence of snapshots showing the implosion of a magnetized star (a) to form a black hole (b). The hole at first inherits the magnetic field from the star. However, the hole has no power to hold on to the field. The field slips off it (c), is converted into electromagnetic radiation, and flies away(d).

If, as we have seen, mountains can be radiated away and magnetic fields can be radiated away, then what is left? What *cannot* be turned into radiation? The answer is simple: Among the laws of physics there is a special set of laws called *conservation laws.* According to these conservation laws, there are certain quantities that can never oscillate or vibrate in a radiative manner, and that therefore can never be converted into radiation and be ejected from a black hole's vicinity. These conserved quantities are the gravitational pull due to the hole's mass, the swirl of space due to the hole's spin (discussed below), and *radially* pointing electric field lines, that is, electric fields that point directly outward (discussed below) due to the hole's electric charge.[5]

Thus, according to Price's theorem, the influences of the hole's mass, spin, and charge are the only things that can remain behind when all the radiation has cleared away. All of the hole's other features will be gone, carried away by the radiation. This means that no measurement one might ever make of the properties of the final hole can possibly reveal any features of the star that imploded to form it, except the star's mass, spin, and charge. From the hole's properties one cannot even discern (according to calculations by James Rartle and Jacob Beken stein, both Wheeler